

MYP 1 Design

Handout 4: Events and User Interaction

Learning Objective

By the end of this lesson, you will be able to:

- Explain what an event is.
- Use different event blocks to start a program.
- Create simple interactive programs.
- Make the sprite respond to user actions.

What is an Event?

An **event** is an action that starts a program.

Events tell the computer **when** to do something.

Examples of Events

- Clicking the Green Flag
- Pressing a key
- Clicking the sprite
- Clicking the Stage

Think of an event like pressing the **Play** button on a TV remote. The TV waits until you press the button before it starts.

Common Event Blocks

Event Block	What it Does
When Green Flag Clicked	Starts the program
When This Sprite Clicked	Starts when the sprite is clicked
When Space Key Pressed	Starts when the Space key is pressed
When Any Key Pressed	Starts when a selected key is pressed

Activity 1: Green Flag Event

Program

When Green Flag Clicked

↓

Say

"Welcome to Coding!"

for 2 seconds

Challenge

Change the message to:

"Hello, everyone!"

Activity 2: Sprite Click Event

Program

When This Sprite Clicked

↓

Say

"You clicked me!"

for 2 seconds

Challenge

Click the sprite several times.

What happens?

Activity 3: Keyboard Event

Program

When Space Key Pressed

↓

Move 50 Steps

Challenge

Change the key to:

- Right Arrow
- Left Arrow
- Up Arrow

What happens when you press each key?

Activity 4: Combining Events

Create two separate programs.

Program 1

When Green Flag Clicked

↓

Say

"Let's Start!"

Program 2

When This Sprite Clicked

↓

Move 100 Steps

Observe

Which action starts first?

What happens when you click the sprite after the program starts?

Activity 5: Create Your Own Interactive Program

Use at least **two different event blocks**.

Your program should:

- Start with the Green Flag.
- Display a welcome message.
- Move when a key is pressed.
- Speak when the sprite is clicked.

Challenge

Can you make your sprite:

- Move using the arrow keys?
- Say different messages when different keys are pressed?

Another important event block is BROADCAST

Broadcast in PictoBlox is a message-passing feature that allows different parts of your program (scripts or sprites) to communicate with each other.

Think of it like:

- One script sending a signal (message)
- Other scripts listening and reacting

Why use Broadcast?

Broadcast is useful when:

- You want to trigger actions in another sprite
- You want to control flow between multiple scripts
- You want to avoid messy long code and keep things organized

How it works

1. Use broadcast [message] → sends a message
2. Use when I receive [message] → receives and reacts

Scenario

When you click a button → a character starts dancing

Code

Script 1 (Button / Main Controller)

when green flag clicked

say "Start Dance!" for 2 seconds

broadcast [dance]

Script 2 (Character Sprite)

when I receive [dance]

repeat 10

 turn 15 degrees

 wait 0.2 seconds

end

What happens?

- Green flag clicked → message "dance" is sent
- Character receives "dance" → starts rotating (dancing)

Another Use Case

Game example:

- Broadcast "game over" → all sprites stop
- Broadcast "level up" → increase difficulty
- Broadcast "start timer" → timer begins

Key Points to Remember

- Broadcast helps in event-driven programming
- Multiple scripts can respond to the same message
- Messages can be custom names (e.g., "jump", "start", "win")

Quick Summary

Event	Purpose
Green Flag	Starts the program
Sprite Clicked	Responds when the sprite is clicked
Key Pressed	Responds when a key is pressed
Events	Tell the program when to begin

Homework Task

Story:

In a magical forest, there are two characters: a Wizard and a Robot. The Wizard has the power to send magical signals. Whenever the Wizard sends a signal, the Robot follows the command.

One day, the Wizard wants the Robot to start dancing when he says “Dance Time”. Later, he wants the Robot to stop when he says “Stop”.

Your job is to create this story using broadcast.

Task Instructions:

Create a project with two sprites:

- Sprite 1: Wizard
- Sprite 2: Robot

What should happen:

- When the green flag is clicked, the Wizard says “Dance Time” and sends a broadcast message
- The Robot receives the message and starts dancing (moving, turning, or changing costumes)
- After a few seconds, the Wizard sends another message like “Stop”
- The Robot receives it and stops dancing

Requirements:

- Use at least two broadcast messages (for example: “dance” and “stop”)
- Use **when I receive** in the Robot sprite
- Add movement or animation for the Robot
- Use wait blocks to control timing

Example flow:

Wizard → says message → broadcasts “dance”

Robot → receives “dance” → starts dancing

Wizard → broadcasts “stop”

Robot → receives “stop” → stops

Challenge Extension:

- Add a third message like “jump” or “spin”
- Add background change when the message is sent
- Add sound effects